Answer Key

Grade 6 Mathematics · Unit 7: Integers and Coordinate Plane · Answers for every item on Sheets 1 and 2 of each lesson packet. Constructed responses show a model answer; accept equivalent reasoning.

LESSON 7-1

Integers and Opposites · Integers in Real Situations

Sheet 1 · Integers and Opposites

1. a. −12 b. 40 c. −9 d. 15 e. −25 f. 7

2. Points at −6 and 6; each is 6 units from 0.

3. A: −7, opposite 7; B: −2, opposite 2; C: 3, opposite −3; D: 9, opposite −9

4. Opposites are on different sides of 0, so the sign changes. −15 is 15 units left of 0; its opposite is 15 units right of 0. The opposite of −15 is 15.

Sheet 2 · Integers in Real Situations

1. a. −35 b. 20 c. −4 d. −8

2. a. 9 b. −15 c. −4

3. A at −3, B at 3, C at −3. A and C are the same point: the opposite of the opposite of −3 is −3.

4. a. noon 6, midnight −6 b. They are opposites: each is 6 units from 0, on different sides.

5. No. The opposite of a negative number is positive. Example: the opposite of −7 is 7, which is positive.

TEST PRACTICE

6. B

7. B

LESSON 7-2

Rational Numbers on a Number Line · Opposites of Rational Numbers

Sheet 1 · Rational Numbers on a Number Line

1. A = −1.75 (−1 3/4) B = −0.5 (−1/2) C = 1.25 (1 1/4)

2. −2.5 halfway between −3 and −2; 1/2 halfway between 0 and 1; −0.75 three fourths left of 0; 1 1/4 one fourth right of 1.

3. a. −5 and −4 b. 3 and 4 c. −1 and 0 d. −3 and −2

4. a. −3.50 b. 2 1/2 c. −4.5

5. a. frog 1.5, fish −2.25 b. the frog (1.5 units from 0 is less than 2.25 units from 0)

6. −1.5 is 1.5 units left of 0, so it is past −1. It belongs halfway between −2 and −1.

Sheet 2 · Opposites of Rational Numbers

1. a. −4.8 b. 2/3 c. 0.05 d. 0 e. 1.6 f. 5/8

2. Q at 2.25 (2 1/4), one fourth to the right of 2.

3. 12.5 points; it is 12.5 points from 0 (same distance as −12.5)

4. a. M = −3.75 (−3 3/4) b. N = 3.75 (3 3/4)

5. −(−2/5) means the opposite of −2/5. The opposite of a negative number is positive, so −(−2/5) = 2/5.

TEST PRACTICE

6. B

7. C

LESSON 7-3

Absolute Value as Distance · Absolute Value as Total Distance

Sheet 1 · Absolute Value as Distance

1. a. 14 b. 8 c. 0 d. 2.5 e. 3/4 f. 1 1/2

2. a. |A| = |−7| = 7 b. |B| = |5| = 5 c. point A

3. a. > (9 > 7) b. = (4 = 4) c. < (−10 < 3) d. = (6.5 = 6.5) e. < (12 < 15) f. < (0 < 1)

4. a. |−35| = 35 and |20| = 20 b. Rosa's (35 > 20)

5. −2, −1, 0, 1, 2 (each is less than 3 units from 0)

6. 4.5 and −4.5. Both are 4.5 units from 0, on different sides.

Sheet 2 · Absolute Value as Total Distance

1. a. 5 + 6 = 11 b. 1 + 6 = 7 c. 5 − 1 = 4

2. a. 10 b. 8 c. 6.5

3. 8 + 15 = 23 degrees

4. −3 and 9 are on opposite sides of 0: 3 + 9 = 12 floors

5. −9 and 5 are on opposite sides of 0, so he should add: 9 + 5 = 14 units.

TEST PRACTICE

6. B

7. B

LESSON 7-4

Comparing Numbers on a Number Line · Ordering Integers and Rational Numbers

Sheet 1 · Comparing Numbers on a Number Line

1. J = −4.5, K = −1, L = 3. a. J < K b. L > K c. J < L

2. a. < b. > c. < d. > e. > f. >

3. Points at −3.5 and −0.5; −3.5 < −0.5

4. a. −5 < −3 (or −3 > −5) b. Malik (−5 is the lower score)

5. Monday was colder: −7 < −2 (or −2 > −7), because −7 is farther left on the number line.

6. Yes. |−6| = 6, and −6 is left of 6 on the number line, so −6 < 6.

Sheet 2 · Ordering Integers and Rational Numbers

1. −3.25, −2.5, −3/4, 1.5, 2 1/4

2. a. −9, −6, 0, 4 b. −3, −0.3, −0.03

3. a. −2 b. −1.04 c. −1/2 d. 0

4. shark (−45.5), dolphin (−30.25), turtle (−12), jellyfish (−8.75)

5. He ordered by distance from 0, as if the numbers were positive, so his list goes greatest to least. For negatives, farther from 0 is less. Correct order: −0.6, −1/2, −0.4.

TEST PRACTICE

6. B

7. B

LESSON 7-5

Plotting Points on the Coordinate Plane · Reflecting Points Across Axes

Sheet 1 · Plotting Points on the Coordinate Plane

Sheet 2 · Reflecting Points Across Axes

1. A (4, 3) I; B (−4, 2) II; C (−2, −4) III; D (3, −4) IV
2. P(2, 4) in I; Q(−4, −1) in III; S(−3.5, 3) in II; G(4, −2.5) in IV. a. the pool b. the garden
3. a. Quadrant III b. Quadrant IV c. on the y-axis d. Quadrant II e. on the x-axis f. Quadrant IV
4. No. Order matters. (−3, 5) is 3 left and 5 up (Quadrant II). (5, −3) is 5 right and 3 down (Quadrant IV).

1. a. B = (−4, −3) b. C = (4, 3)
2. a. (6, −2) b. (4, −7)
3. (7, 2.5)
4. K = (−6, 5). Reflecting across the x-axis flips the sign of y, so flip it back: −5 becomes 5.
5. He changed the y-coordinate. Across the y-axis, y stays the same and x changes sign. Correct point: (3, −8).

TEST PRACTICE

6. B
7. A

LESSON 7-6

Distance Between Points on a Grid · Comparing Distances on the Plane

Sheet 1 · Distance Between Points on a Grid

1. A(−4, 3), B(3, 3), C(3, −2), D(−4, −4). a. 4 + 3 = 7 units b. 3 + 2 = 5 units c. 3 + 4 = 7 units
2. a. 8 b. 5 c. 9 d. 11 e. 5 f. 6.5 g. 10 h. 4.5
3. Same x; 3 + 5 = 8 blocks
4. Q = (−4, −3)
5. Both y-coordinates are negative (same side of the x-axis), so subtract: 6 − 2 = 4 units.

Sheet 2 · Comparing Distances on the Plane

1. H(−4, 2), L(3, 2), P(−4, −3). a. 4 + 3 = 7 blocks b. 2 + 3 = 5 blocks c. the Park, by 2 blocks
2. a. 8 − 3 = 5 b. 6 + 7 = 13 c. 7 − 1 = 6 d. 2.5 + 4.5 = 7
3. Ana: 5 + 6 = 11 units. Jay: 4 + 6 = 10 units. Ana runs 1 unit farther.
4. Add. The y-coordinates 7 and −2 have different signs, so the path crosses the x-axis: 7 + 2 = 9 units.

TEST PRACTICE

5. A – JK = 4 + 6 = 10, MN = 3 + 5 = 8
6. C

LESSON 7-7

Polygons and Perimeter on the Plane · Designing Polygons with Scale

Sheet 1 · Polygons and Perimeter on the Plane

1. A(−4, 3), B(2, 3), C(2, −2), D(−4, −2). a. 4 + 2 = 6 units b. 3 + 2 = 5 units c. 6 + 5 + 6 + 5 = 22 units d. 6 × 5 = 30 square units
2. a. 3 + 4 = 7 units b. 2 + 4 = 6 units c. ½ × 7 × 6 = 21 square units
3. Sides 5 + 6 = 11 m and 3 + 4 = 7 m; P = 11 + 7 + 11 + 7 = 36 m
4. Each side is 3 + 2 = 5 units; P = 4 × 5 = 20 units
5. (−2, −1). It must line up with (−2, 5) vertically (x = −2) and with (3, −1) horizontally (y = −1).

Sheet 2 · Designing Polygons with Scale

1. A(−3, 4), B(−3, −2). a. 4 + 2 = 6 units b. 6 × 5 = 30 ft c. 1,200 ÷ 30 = 40 ft d. 40 ÷ 5 = 8 units
2. a. 28 m b. 9 units
3. 14 ft × 10 ft = 140 ft²
4. Grid sides 2 + 2 = 4 units (12 ft) and 4 + 1 = 5 units (15 ft). P = 12 + 15 + 12 + 15 = 54 ft
5. Lily subtracted 5 − 4. The signs differ, so add: 4 + 5 = 9 units; 9 × 3 = 27 ft.

TEST PRACTICE

6. B
7. A